**IILM UNIVERSITY, GREATER NOIDA**

**SCHOOL OF COMPUTER SCIENCE & ENGINEERING**

# DATASET AND RESULTS REPORT

**Project Title:** Predictive Maintenance System for Data Centre Servers

## 1. Project Details

| | |
|---|---|
| **Project ID** | 26P_3CSE276-Predictive Maintenance System for Data Centre Servers |
| **Project Team Members** | Richa Goswami (2410031226)<br>Riya Tyagi (2410031227)<br>Gungun Baliyan (2410031220)<br>Divyansh Singh (2410031269) |
| **Project Guide** | Shubham Jayasval |

## 2. Dataset Description

**Note:** The dataset used is synthetic (, not real data-centre telemetry. All figures below were measured by running the code on that dataset; results on real server data will differ.

| | |
|---|---|
| **Dataset name** | data_Q1_2026 |
| **Source** | BlackBlaze |
| **Number of records** | 5000 |
| **Number of features** | 4 input features, 1 target |
| **Target / label column** | maintenance_needed |
| **Class distribution** | 3,270 normal (65.4%), 1,730 maintenance needed (34.6%) |
| **Missing values** | 0 |

## 3. Features

| Feature | Description | Unit | Valid Range |
|---|---|---|---|
| **cpu_usage** | CPU utilisation | Percent (%) | 0 – 100 |
| **memory_usage** | Memory utilisation | Percent (%) | 0 – 100 |
| **temperature** | Server temperature | Degrees Celsius | 10 – 110 |
| **disk_usage** | Disk utilisation | Percent (%) | 0 – 100 |

## 4. Feature Statistics

| Feature | Mean | Std Dev | Min | Median | Max |
|---|---|---|---|---|---|
| **cpu_usage** | 54.53 | 19.71 | 0 | 54.92 | 100 |
| **memory_usage** | 59.9 | 17.98 | 0 | 59.64 | 100 |
| **temperature** | 49.87 | 7.2 | 24.59 | 49.72 | 77.06 |
| **disk_usage** | 54.76 | 19.71 | 0 | 54.36 | 100 |

## 5. Data Preprocessing

- Missing values: none in the dataset; at prediction time any missing field is rejected with an error.
- Invalid values: non-numeric or out-of-range inputs are rejected by the validation step.
- Scaling: standardisation applied only for Logistic Regression; tree models use raw values.
- Feature engineering: none; the four monitored metrics are used directly.
- How the data was built: temperature depends on CPU and memory load plus noise; the maintenance label comes from a weighted risk score of the four metrics plus random noise.

## 6. Results Summary

| Item | Result | Remarks |
|---|---|---|
| **Records used for training** | 4000 | 80% stratified split |
| **Records used for testing** | 1000 | 20% stratified split |
| **Best model** | Gradient Boosting | Highest F1 on test set |
| **Test accuracy** | 87.70% | |
| **Test F1-score** | 0.817 | |
| **Maintenance servers caught / total** | 275 / 346 | Recall 0.795 |
| **False alarms / normal servers** | 52 / 654 | |

## 7. Sample Outputs

| CPU % | RAM % | Temp (°C) | Disk % | Predicted Status | Probability |
|---|---|---|---|---|---|
| **30** | 40 | 40 | 35 | NORMAL | 0.0051 |
| **55** | 60 | 50 | 50 | NORMAL | 0.1203 |
| **75** | 70 | 60 | 60 | MAINTENANCE ALERT | 0.9801 |
| **90** | 85 | 80 | 70 | MAINTENANCE ALERT | 0.9914 |

| CPU % | RAM % | Temp (°C) | Disk % | Predicted Status | Probability |
|---|---|---|---|---|---|
| **98** | 95 | 95 | 95 | MAINTENANCE ALERT | 0.9926 |
| **20** | 30 | 35 | 90 | NORMAL | 0.0214 |

## 8. Conclusion

The dataset has 5,000 complete records with no missing values and a 35% maintenance class. Gradient Boosting reached 87.70% accuracy and an F1-score of 0.817 on the held-out test set. Because the data is synthetic, these results demonstrate that the pipeline works end to end; they are not evidence of real-world accuracy.